

Feedback — Group 7, Music Recommendation with GNN and KG

Part A — Feedback to students

1. Overall assessment

The project is technically on track and has moved well beyond the design stage. You now have a substantial graph-based recommendation pipeline, several baselines, learning curves, ablations, statistical tests, and member contributions. The evidence is promising, but the main model comparison is weakened by the directed/undirected HGT evaluation mismatch. Before the poster and final report, the most important fix is to present a clean, consistent final comparison where the HGT result is evaluated under the same assumptions used at inference.

2. What is already good

- The pipeline is ambitious and mostly implemented: feature extraction, KG construction, KGE pretraining, HGT training, collaborative filtering, and XGBoost reranking.
- The report includes relevant recommendation metrics at K=10, including ranking quality, coverage, and popularity bias.
- The HGT ablations are useful: random versus RotatE initialization, and the logit-adjustment sweep.
- Learning curves are included for the autoencoder, KGE models, XGBoost, and HGT.

3. Main issues to fix before the final delivery

1. **Issue:** The HGT benchmark row is affected by a directed/undirected evaluation mismatch.
Why it matters: This makes the central GNN result hard to interpret and may unfairly understate or overstate the method.
Concrete action: Add one final table where HGT is evaluated consistently. Either retrain/evaluate with directed edges or clearly use the self-consistent undirected inference protocol, but do not leave the main row as a known artefact.
2. **Issue:** KNN-CF currently has the best overall score, while the proposed HGT contribution is less clearly positioned.
Why it matters: The final scientific claim should explain what the graph model adds beyond a strong recommender baseline.
Concrete action: Add a short paragraph comparing HGT versus KNN-CF in terms of ranking quality, coverage, diversity, cold-start potential, and interpretability.
3. **Issue:** The XGBoost candidate bottleneck is diagnosed but not made visually obvious.

Why it matters: Readers may interpret low XGBoost NDCG as model failure rather than retrieval limitation.

Concrete action: Add a compact pipeline figure or table showing candidate generation size, reranker input, and the resulting ceiling.

4. **Issue:** The attention/explainability claim is stated but not demonstrated with examples.

Why it matters: Interpretability is one of the main advantages of a KG/GNN recommender.

Concrete action: Add 2–3 recommendation examples with top contributing edge types or neighboring entities.

5. **Issue:** The final model selection criterion is complex.

Why it matters: The composite score mixes ranking, coverage, and popularity, so the reader needs to know exactly why the chosen checkpoint is the final model.

Concrete action: State the validation checkpoint criterion once in the experimental setup and reuse the same criterion in all final tables.

4. Milestone 2 requirements checklist

- Training methods: **OK**.
- Experiments: **OK**.
- Results: **OK**.
- Learning curves: **OK**.
- Contribution of each group member: **OK**.

5. Suggestions from M1 that still need attention

The evaluation protocol still needs attention because of the HGT directed/undirected mismatch; the error analysis should be strengthened with recommendation-level examples.

6. Recommended final experiments or analyses

Must do: - Recompute or restate the final HGT comparison under a self-consistent inference protocol. - Add a short qualitative explanation example for HGT recommendations.

Nice to have: - Run inference with a larger candidate pool for XGBoost if this can be done without retraining. - Add a small cold-start or low-history user slice analysis if the data already supports it.

7. What to include in the final report

- **Methodology:** KG schema, node/edge types, feature construction, AE/KGE/HGT pipeline, and baseline definitions.

- **Experimental setup:** splits, metrics, checkpoint selection, hyperparameters, hardware/training time, and statistical test setup.
- **Results:** one main table with consistent evaluation, HGT ablations, learning curves, and baseline comparison.
- **Discussion/limitations:** directed/undirected issue if not fully fixed, popularity bias, candidate retrieval bottleneck, and static implicit-feedback limitations.
- **Member contributions:** keep the current concrete breakdown.

8. What to include in the poster

Show the pipeline visually. Include the dataset/KG summary, a compact main result table, the HGT ablation table, one learning curve panel, one recommendation explanation example, 2–3 key takeaways, limitations, and the member contribution statement. The poster should make the ranking-versus-diversity trade-off visually obvious.